

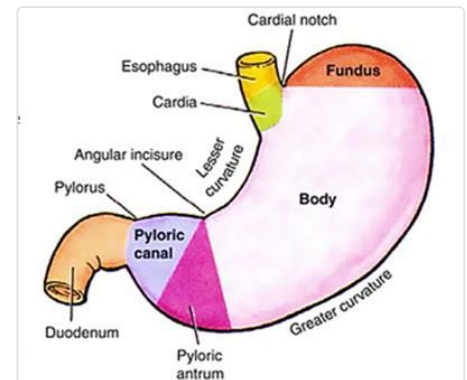
**BEFORE YOU READ — ANSWER FROM MEMORY**

1. At which vertebral levels do the cardiac and pyloric openings of the stomach lie?
2. A posterior duodenal ulcer bleeds from which artery — and a posterior gastric ulcer from which?
3. Name three features that tell the jejunum from the ileum at surgery.

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Stomach — position, shape and parts

Location	Left hypochondriac, epigastric and umbilical regions; T10 → L3
Shape	J-shaped when empty; horizontal (“teapot”) in obese people
Size	~10 in long; ~1000 ml at puberty → 1500–2000 ml in adults
Interior	Rugae (longitudinal gastric folds), most marked toward the pyloric part along the greater curvature
Plan	2 openings · 2 curvatures · 2 surfaces · 4 parts



Parts of the stomach: cardia, fundus, body, pyloric antrum and canal.

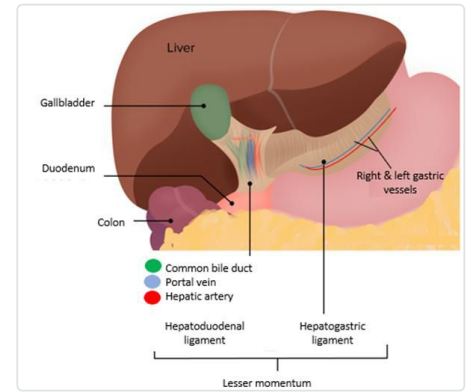
Feature	Key facts
Cardiac opening	Continuous with oesophagus · behind left 7th costal cartilage , 1 in from sternum · T11
Pyloric opening	Continuous with duodenum · L1 (transpyloric plane) , ~1.5 cm right of midline · guarded by the pyloric sphincter (“gatekeeper”)
Greater curvature	Convex, left border, 5x longer than lesser · attaches greater omentum · cardiac notch at upper end
Lesser curvature	Concave, right border · attaches lesser omentum · incisura angularis at lower end
Part	Where / what
Cardia	Left of midline at T11 , just below the xiphisternal joint
Fundus	Dome under the left dome of the diaphragm ; 5th left rib, MCL (per slides); the gas bubble on X-ray
Body	Largest part; fundus → incisura angularis
Pyloric part	Pyloric antrum (wide, proximal) + pyloric canal (narrow) → pylorus; right of midline at L1

2 The omenta

	Greater omentum (“apron”)	Lesser omentum
Extent	Greater curvature + first inch of duodenum → transverse colon and transverse mesocolon	Lesser curvature, right side of abdominal oesophagus, first 2 cm of duodenum → liver (inverted L: fissure for ligamentum venosum + porta hepatis)
Contents	Fat · milky spots (macrophage patches) · right & left gastroepiploic vessels	See free margin and hepatogastric ligament below
Clinical	“Policeman of the abdomen” : moves to infection and seals it off; fat store; surgical graft material	Free margin = hepatoduodenal ligament (portal triad)

Lesser omentum contents (superficial → deep)

Free margin	Bile duct — right, anterior Hepatic artery proper — left, anterior Portal vein — posterior, deepest
Hepatogastric ligament	Right gastric vessels → left gastric vessels → lymphatics & nodes → vagal branches → sympathetic fibres (celiac plexus)

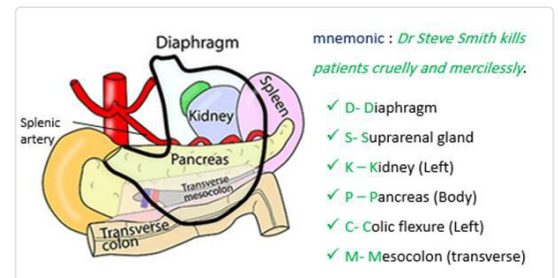


Lesser omentum: hepatoduodenal (free margin) and hepatogastric parts.

3 Surfaces and the stomach bed

Anterior (antero-superior)	Anterior abdominal wall, left costal margin, left pleura & lung, diaphragm, left lobe of liver
Posterior (postero-inferior)	Lesser sac, diaphragm and the stomach bed : spleen, left suprarenal, upper left kidney, splenic artery , pancreas , transverse mesocolon, transverse colon

Trap: the liver is an **anterior** relation — it is never part of the stomach bed.



Stomach bed mnemonic from the lecture: "Dr Steve Smith kills patients cruelly and mercilessly."

4 Stomach — vessels, lymph and nerves

Artery	Arises from	Supplies
Left gastric	Celiac trunk	Lower 1/3 of oesophagus + upper right part of stomach
Right gastric	Hepatic artery	Lower right part
Short gastrics	Splenic artery	Fundus
Left gastroepiploic	Splenic artery	Upper greater curvature
Right gastroepiploic	Gastrooduodenal (of hepatic)	Lower greater curvature

Veins (all → portal circulation)	Drain into
Left & right gastric	Portal vein directly
Short gastrics + left gastroepiploic	Splenic vein
Right gastroepiploic	Superior mesenteric vein

Lymph	Follows the arteries → gastric, gastroepiploic and short gastric nodes → all ends in the celiac nodes (root of celiac artery)
Sympathetic	T6-T9 via greater splanchnic nerves and celiac plexus
Parasympathetic	Anterior vagal trunk = left vagus → anterior surface · Posterior vagal trunk = right vagus → mainly posterior surface
Collaterals	Rich arterial anastomoses → one or more arteries can be ligated in partial/total gastrectomy without endangering the remnant. Antrectomy: omentum incised parallel and inferior to the right gastro-omental artery , ligating all its omental branches.

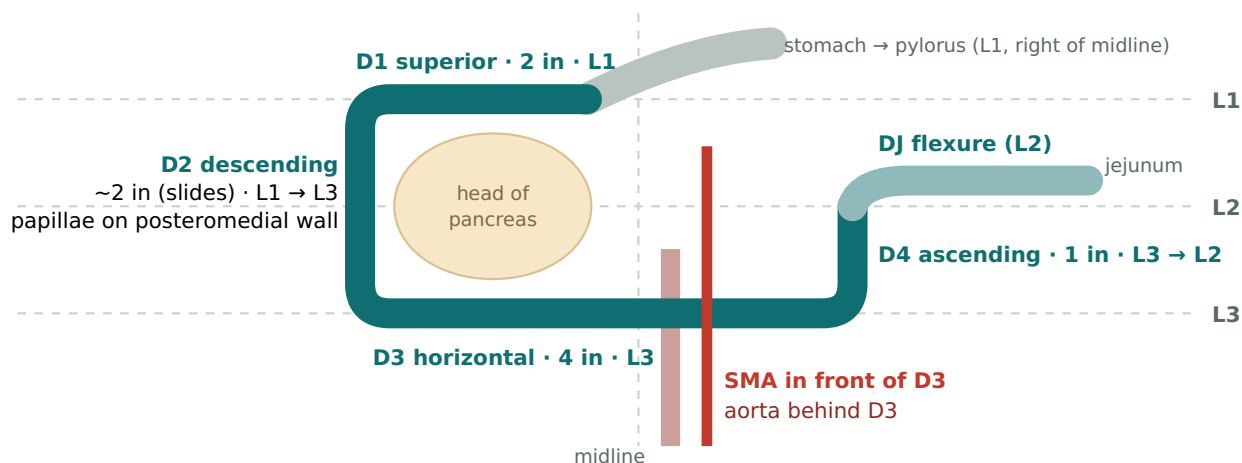
5 Stomach — clinical anatomy

Vagotomy	Vagal trunks cut as they enter the abdomen → less parietal-cell acid → treats peptic ulcer
Gastric pain	Felt in the epigastrium — T6–T9 also supply the upper abdominal wall
Gastric ulcer	Usually on the lesser curvature : thin epithelium, reduced blood supply with fewer anastomoses, gastric canal receives chyme, shorter length
Posterior ulcer	Erodes into pancreas → pain referred to back; erosion of splenic artery → severe haemorrhage
Anterior ulcer	Contents escape into greater sac → may reach the right iliac fossa → mimics appendicitis
Gastric carcinoma	Often pyloric antrum (greater curvature per slides) · lymphatic spread via thoracic duct → left supraclavicular node = Virchow's node ; palpable = Troisier's sign
Surgery / hernia	Hiatus hernia (stomach through diaphragm → reflux, dysphagia) · sleeve gastrectomy (greater curvature removed, tube along lesser curvature) · gastric bypass (small pouch + bypassed small bowel)

6 Small intestine and the duodenum

Small intestine	Longest part of the gut, ~ 20 ft (6 m) ; does 90% of digestion and absorption; duodenum + jejunum + ileum
Duodenum	First, shortest, widest, most fixed part · “12 fingers” · C-shaped around the head of pancreas · 10 in (25 cm) · umbilical region, L1–L3
Peritoneum	Retroperitoneal except the proximal 2.5 cm (held by lesser omentum above, greater omentum below)

THE DUODENAL C-LOOP — parts, lengths, levels (anterior view: patient's right on the left)



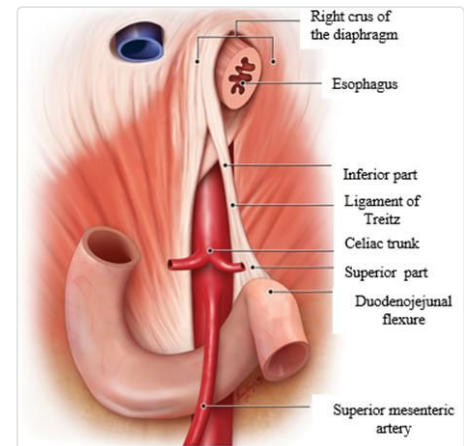
Total ≈ 10 in (25 cm). Upper half → celiac route (superior pancreaticoduodenal). Lower half → SMA route (inferior pancreaticoduodenal).

Part	Length · level	Distinguishing facts	Clinical
1st (superior)	2 in · L1	1st inch intraperitoneal, 2nd inch retroperitoneal · only part with end arteries · only part without circular folds	Commonest ulcer site. Anterior perforation → greater sac above transverse colon → right paracolic gutter → RIF (mimics appendicitis). Posterior ulcer (65%) → erodes gastroduodenal artery
2nd (descending)	~2 in · L1–L3	Posteromedial wall: major papilla (bile + main pancreatic duct, ampulla of Vater); minor papilla ~2 cm above it (accessory duct of Santorini)	Congenital diverticula on the medial wall · most protected part (paravertebral gutter)
3rd (horizontal)	4 in · L3	Runs left across lower L3 body, between SMA (front) and aorta (behind)	Most vulnerable to external injury · SMA syndrome : young patient, rapid weight loss, post-prandial pain, nausea, vomiting

4th (ascending)	1 in · L3→L2	Ends at the duodenojejunal flexure	Gallstone may ulcerate from gallbladder into duodenum; colon / right kidney surgery can damage it
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7 Ligament of Treitz and duodenal vessels

What	Fibromuscular band suspending the DJ flexure from the right crus of the diaphragm	
Why it matters	Stops the flexure being pulled down by the small-bowel loops · landmark between upper and lower GI tract · may kink the flexure → partial obstruction	
	Upper half	Lower half
Artery	Superior pancreaticoduodenal (gastroduodenal ← common hepatic ← celiac)	Inferior pancreaticoduodenal (SMA)
Vein	→ portal vein (via gastroduodenal or splenic veins)	→ superior mesenteric vein
Lymph	Pancreaticoduodenal → gastroduodenal → celiac nodes	Pancreaticoduodenal → superior mesenteric nodes

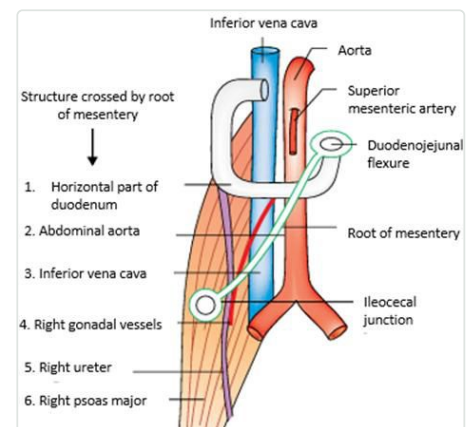


Ligament of Treitz running from the right crus to the duodenojejunal flexure.

8 Jejunum, ileum and the mesentery

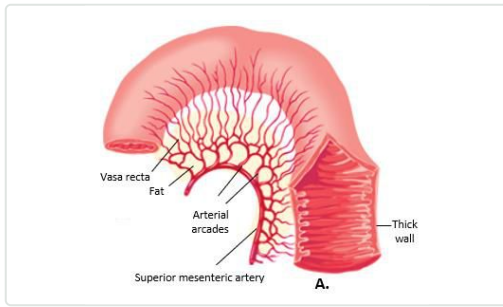
Duodenojejunal flexure → ileocecal junction, ~20 ft (6 m): **upper 2/3 jejunum, lower 1/3 ileum**, both suspended by the mesentery.

Mesentery	Fan-shaped fold with a root and a ~6 m free (intestinal) border that pleats around the bowel and allows mobility; the root divides the infracolic compartment into right and left
Root line	DJ flexure (left side of L2) → ileocecal junction (right sacroiliac joint)
Root crosses	From above down: 3rd part of duodenum → aorta → IVC → right gonadal vessels → right ureter → right psoas major
Contains	SMA and SMV, lymphatics and nodes, fat, nerve fibres

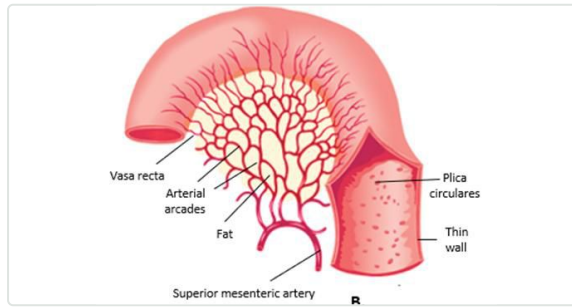


Structures crossed by the root of the mesentery, in order.

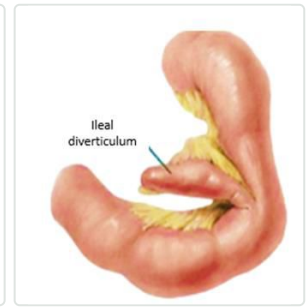
Feature	Jejunum	Ileum
Location	Left hypochondriac region, middle of small intestine (per slides)	Right iliac region, distal small intestine
Length	~2.5 m	~3.5 m
Diameter / wall	Wider, thicker , more vascular	Narrower, thinner, less vascular
Colour	Deeper red	Paler pink
Circular folds	Prominent and dense	Smaller and sparse
Lymphoid tissue	Few nodes	Numerous Peyer's patches
Arcades / vasa recta	Few arcades, long vasa recta	Multiple arcades, short vasa recta
Mesenteric fat	Less fat near wall → translucent "white windows"	Fat reaches the wall → no windows



Jejunum: few arcades, long vasa recta, thick wall.



Ileum: multiple arcades, short vasa recta, fat to the wall.



Meckel's (ileal) diverticulum.

Arteries	12-15 jejunal and ileal branches of the SMA → arcades → vasa recta to the mesenteric border · terminal ileum: ileal branches of the ileocolic artery
Veins	Follow the SMA → SMV → portal vein (carries protein and carbohydrate products to the liver)
Lymph	Mesenteric nodes → superior mesenteric nodes at the SMA origin
Meckel's diverticulum	Remnant of the proximal vitellointestinal duct · “rule of 2s”: 2% of people, 2 in (5 cm) long, 2 ft (60 cm) proximal to the ileocecal junction · more common in men · wall like ileum

P Must-remember pearls

- **Cardiac opening T11** (behind left 7th costal cartilage) · **pyloric opening L1**, transpyloric plane, 1.5 cm right of midline.
- Free margin of lesser omentum: **bile duct (ant-right)**, **hepatic artery proper (ant-left)**, **portal vein (posterior, deepest)**.
- Stomach bed = spleen, left suprarenal, upper left kidney, splenic artery, pancreas, transverse mesocolon, transverse colon — **not the liver**.
- All gastric lymph ends in the **celiac nodes**; gastric cancer reaches **Virchow's node** via the **thoracic duct** (Troisier's sign).
- Anterior vagal trunk = **left** vagus; posterior vagal trunk = **right** vagus.
- Posterior D1 ulcer → **gastroduodenal artery**; posterior gastric ulcer → **splenic artery** (+ back pain from pancreas).
- D1: only end arteries, no circular folds · D2: papillae, most protected · D3: SMA syndrome, most vulnerable · D4: to DJ flexure at L2.
- Ligament of Treitz: **right crus → DJ flexure**; upper/lower GI boundary.
- Root of mesentery: L2 left → right sacroiliac joint; crosses D3, aorta, IVC, right gonadal vessels, right ureter, right psoas.
- Jejunum: wide, thick, red, few arcades, long vasa recta, white windows · Ileum: Peyer's patches, many arcades, short vasa recta.
- Meckel's: 2% · 2 in · 2 ft from the ileocecal junction · vitellointestinal duct remnant.

Q Test yourself

1. A 48-year-old man with chronic epigastric pain now has pain that radiates to his back and a sudden large haematemesis. Endoscopy shows a posterior gastric ulcer penetrating the pancreas. Which artery has most likely eroded?

- A. Gastroduodenal artery
- B. Right gastric artery
- C. Splenic artery
- D. Superior mesenteric artery
- E. Right gastroepiploic artery

2. A 21-year-old woman lost a lot of weight after a long illness. She now vomits after every meal. CT shows the duodenum dilated up to the point where it crosses L3. What is compressing it?

- A. Ligament of Treitz kinking the DJ flexure
- B. Superior mesenteric artery against the aorta
- C. Head of pancreas
- D. Gallbladder
- E. Root of the transverse mesocolon

3. At laparotomy a surgeon finds a 5 cm blind pouch arising from the small bowel about 60 cm proximal to the ileocecal junction. From which embryonic structure is this derived?

- A. Allantois
- B. Urachus
- C. Vitellointestinal duct
- D. Ventral mesogastrium
- E. Dorsal pancreatic bud

4. A surgeon places a clamp across the free margin of the lesser omentum. Which structure lies deepest in the clamp?

- A. Bile duct
- B. Hepatic artery proper
- C. Right gastric artery
- D. Portal vein
- E. Left gastric vein

5. A loop of bowel at surgery is narrow, thin-walled and pale; its mesentery has several tiers of arcades, short vasa recta and fat reaching the wall. Which lymphoid feature would you expect in this segment?

- A. Numerous Peyer's patches
- B. No lymphoid tissue at all
- C. Milky spots
- D. Few scattered lymph nodes only
- E. Brunner's glands

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 C — A posterior gastric ulcer can penetrate the pancreas (back pain) and erode the splenic artery. The gastroduodenal artery is the one at risk from a posterior duodenal ulcer.

2 B — SMA syndrome: the 3rd part is compressed between the SMA in front and the aorta behind. Suspect it after rapid weight loss.

3 C — Meckel's (ileal) diverticulum is a remnant of the vitellointestinal duct: 2%, 2 in, 2 ft from the ileocecal junction.

4 D — Portal vein is posterior and deepest; the bile duct lies anterior-right and the hepatic artery proper anterior-left.

5 A — These are ileal features, and the ileum has numerous Peyer's patches. Milky spots belong to the greater omentum.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Anatomy GIT Block · MEDucation by Mattin Majid · GIT Anatomy SGL 3 slides, 2026-2027
Study alongside the lecture, not instead of it.